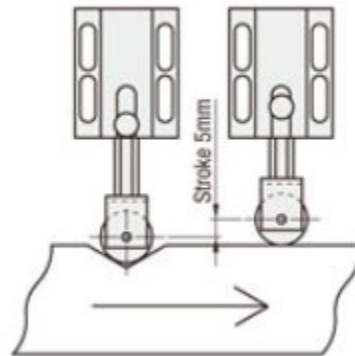
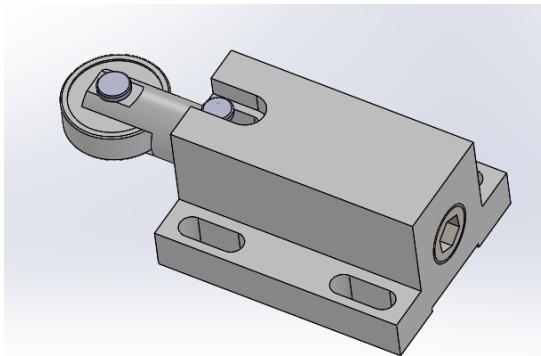




Assignment:	Roller Plunger Design	Value: 3% Bonus marks
Submission Date: September 15,2026	Group:	
Name	Student number:	

1- Assignment Overview

- The preliminary roller plunger design shown below is intended for use in a machine door assembly to provide a snap-action or detent-type function where sliding door positioning is required.
- The objective of this assignment is to evaluate the preliminary design and develop an improved version that is simpler to manufacture, faster and easier to assemble, cost-effective, and functionally reliable.
- Students are expected to apply engineering design principles, including material selection, dimensional design, fits and tolerances, manufacturability, assembly considerations, and mechanical performance.



2- Preliminary Design Specifications

The following specifications shall be considered the starting point for the redesign.

Main Body Material	Aluminum 6061-T6
Main Body Surface Treatment	Clear Anodize
Pin Material	AISI 1045 Carbon Steel
Pin Surface Treatment	Electroless Nickel Plating
Roller Material	AISI 416 Stainless Steel, or other suitable material

Roller Surface Treatment	Electroless Nickel Plating
Roller Heat Treatment	49–51 HRC
Plunger Stroke	0.250 in (6.35 mm)
Initial Load	20 N
Maximum Load	40 N
Roller Diameter	0.750 in (19.05 mm)
Roller Width	0.1875 in (4.76 mm)

Where an alternative material or manufacturing process is proposed, the student must provide a brief engineering justification.

3- Design Objectives

Redesign the preliminary roller plunger assembly to achieve improvements in the following areas:

A. Design for Manufacturing and Assembly (DFMA)

The redesign should demonstrate consideration of **Design for Manufacturing and Assembly (DFMA)** principles.

Compare the preliminary design with the proposed design and identify specific improvements.

Develop a design that:

- Reduces manufacturing time.
- Reduces the number of machining operations.
- Minimizes complex or unnecessary features.
- Reduces material waste where practical.
- Uses readily available materials and standard components where possible.
- Can be manufactured using conventional machining processes.
- Reduces overall manufacturing cost.

B. Design for Assembly

Improve the assembly process by:

- Reducing the total number of components where practical.
- Reducing the number of assembly operations.
- Making components easier to orient and install.
- Eliminating unnecessary fasteners or specialized assembly operations.
- Considering ease of maintenance and replacement.
- Minimizing the required assembly time.

C. Performance and Functionality

The redesigned assembly must:

- Provide a plunger stroke of approximately **6.35 mm (0.250 in)**.
- Provide an initial operating load of approximately **20 N**.
- Withstand a maximum operating load of **40 N**.

- Provide smooth and reliable roller movement.
- Minimize friction and wear between contacting components.
- Maintain adequate alignment during operation.
- Provide sufficient strength and stiffness for the intended application.
- Be suitable for repeated machine operation.

Any deviation from the preliminary specifications must be clearly identified and justified.

4- Assignment Requirements

Produce a complete set of engineering drawings suitable for manufacturing and assembly.

The drawing package shall include:

A. Main Assembly Drawing

The assembly drawing shall include:

- Complete assembled view
- Appropriate section view(s), if needed
- Overall dimensions where appropriate
- Component identification balloons
- Assembly notes

B. Bill of Materials (BOM)

Provide a complete BOM identifying all manufactured and purchased components.

C. Detailed Manufacturing Drawings

Provide detailed drawings for **manufactured parts only**.

Each manufacturing drawing shall contain sufficient information to manufacture and inspect the part, including:

- All required dimensions
- General tolerances, limits and fits
- Surface finish requirements
- Material specification
- Heat treatment
- Surface treatment/coating
- Manufacturing notes
- Quantity

D. Fits and Tolerances

Show appropriate fits and tolerances on the drawings wherever required.

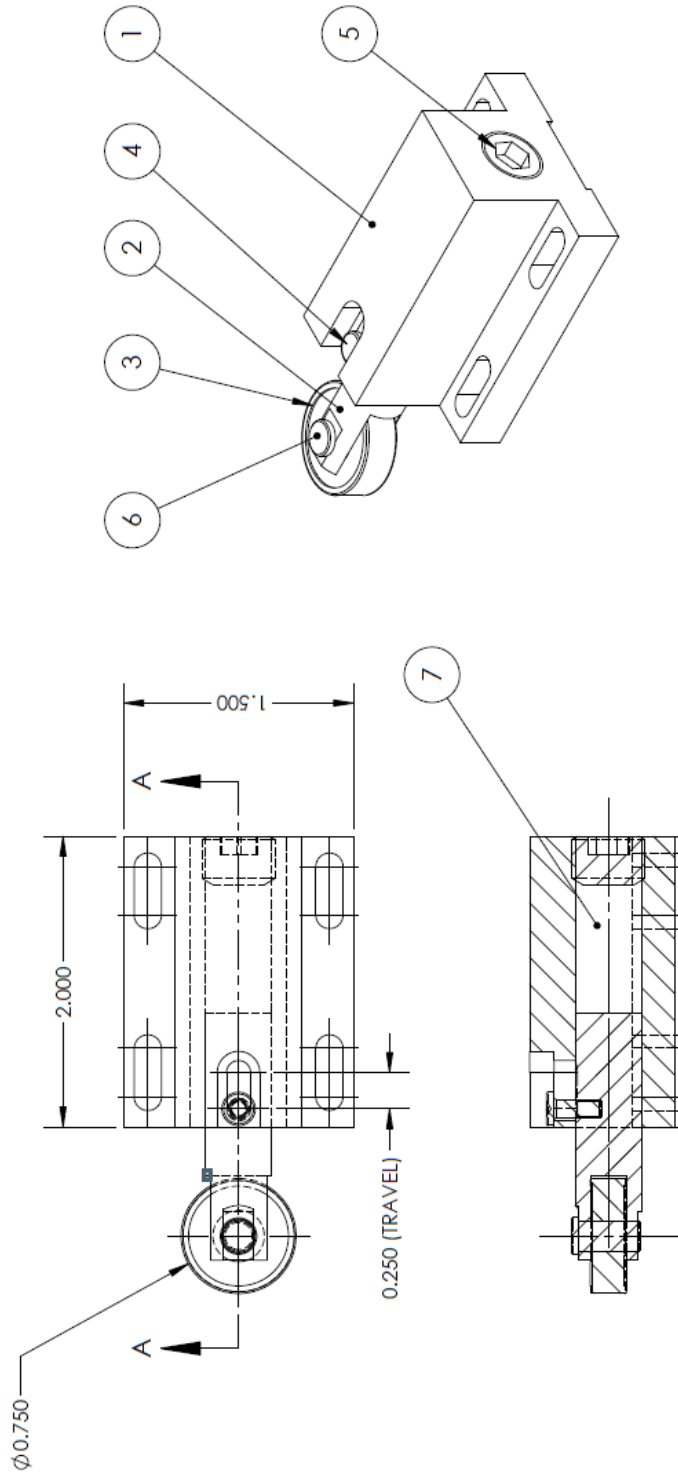
Students are expected to determine suitable fits based on the intended function rather than applying tolerances arbitrarily.

Particular attention should be given to:

- Pin-to-body interface
- Sliding/moving surfaces
- Roller-to-pin retention
- Sliding-limit retention and pin
- Assembly clearances

Where a standard ISO or ANSI fit is selected, identify the selected fit and provide a brief justification.

ITEM NO.	DESCRIPTION	SUPPLIER/ MTR'L	PRT# / PRT #	QTY.
1			DWG # 02-ASSI-00	1
2			DWG # 03-ASSI-00	
3		Mcmaster Carr	60355K505, 1.125X 0.375 X 0.250	1
4			Ø0.25 X 1.0"Lg	1
5	SET SCREW	STOCK	M12X1.75 X 14mm Lg	1
6	PIN			1
7	SPRING			1



SECTION A-A

UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. DO NOT SCALE DRAWING
3. BREAK SHARP EDGES
4. MACHINED SURFACES
5. TOLERANCES:
.X ± .020
.XX ± .010
.XXX ± .005
FRACTIONAL 1/22
ANGULAR ± 0.15°

HUMBER
Humber College

DATE SEP 21 2025 DWG # 01-ASSI-00
ASSI NAME # ROLLER PLUNGER
STUDENT #
DRAWN BY COMP. MACH.DESIGN SCALE 1:1